Assignment 4: Schema creation and constraints

Name: Arnav Singhal

MIS: 612203012

Div: 1; **Batch**: T1

Statement: Create the schema and constraints on the relations.

Modify the trains schema which we saw earlier to create constraints to check the following:

- The value of timein is always less than or equal to timeout

  Query:

  ```
  ALTER TABLE trainhalts
  ADD CONSTRAINT chk_time_order
  CHECK (timein IS NULL OR timeout IS NULL OR timein <= timeout);
  ```

- When a train is removed from service, all its halts should be deleted.

  Query:

  ```
  ALTER TABLE trainhalts
  ADD CONSTRAINT fk_train
  FOREIGN KEY (id) REFERENCES train(id)
  ON DELETE CASCADE;
  ```

- Insert inconsistent data and verify the constraints.

  Queries and Output:

  ```
  INSERT INTO train VALUES ('T1', 'TEST-EXPRESS');

  -- This query fails as timein > timeout
  INSERT INTO trainhalts VALUES
  ('T1', 0, 'CST', NULL, '16.00'),
  ('T1', 1, 'BYC', '16.15', '16.10');

  -- Corrected version of the above query
  INSERT INTO trainhalts VALUES
  ('T1', 0, 'CST', NULL, '16.00'),
  ('T1', 1, 'BYC', '16.10', '16.15');
  ```

```
-- Displaying all records in trainhalts corresponding to id = ‘T1’
     SELECT * FROM trainhalts WHERE id = ‘T1’;
```

Output:

```
+------+-------+--------+--------+---------+
| id   | seqno | stcode | timein | timeout |
+------+-------+--------+--------+---------+
| T1   |     0 | CST    | NULL   | 16.00   |
| T1   |     1 | BYC    | 16.10  | 16.15   |
+------+-------+--------+--------+---------+
```

```
     -- Deleting the newly added train
     DELETE FROM train WHERE id = ‘T1’;

     -- All halts for train with id=‘T1’ have been automatically deleted
     SELECT * FROM trainhalts;
```

Output:

```
Empty set (0.00 sec)
```

Write SQL Create table statements to create the following schema. Include all appropriate primary and foreign key declarations. Choose appropriate types for each attribute.

- remotecentre(centreId, college, town, state)

Creating table:

```
CREATE TABLE remotecentre (
      centreId    INT               PRIMARY KEY,
      college     VARCHAR(50)       NOT NULL,
      town        VARCHAR(50)       NOT NULL,
      state       VARCHAR(50)       NOT NULL
);
```

Inserting data:

```
INSERT INTO remotecentre VALUES
(1, 'COEP Technological University', 'Pune', 'Maharashtra'),
(2, 'IIT Madras', 'Chennai', 'Tamil Nadu'),
(3, 'NIT Trichy', 'Tiruchirappalli', 'Tamil Nadu'),
(4, 'BITS Pilani', 'Pilani', 'Rajasthan'),
(5, 'IIT Kharagpur', 'Kharagpur', 'West Bengal'),
(6, 'Delhi Technological University', 'New Delhi', 'Delhi'),
(7, 'VIT University', 'Vellore', 'Tamil Nadu'),
(8, 'IIIT Hyderabad', 'Hyderabad', 'Telangana'),
(9, 'IISc Bangalore', 'Bengaluru', 'Karnataka'),
(10, 'IIT Guwahati', 'Guwahati', 'Assam');
```

Table contents:

centreId	college	town	state
1	COEP Technological University	Pune	Maharashtra
2	IIT Madras	Chennai	Tamil Nadu
3	NIT Trichy	Tiruchirappalli	Tamil Nadu
4	BITS Pilani	Pilani	Rajasthan
5	IIT Kharagpur	Kharagpur	West Bengal
6	Delhi Technological University	New Delhi	Delhi
7	VIT University	Vellore	Tamil Nadu
8	IIIT Hyderabad	Hyderabad	Telangana
9	IISc Bangalore	Bengaluru	Karnataka
10	IIT Guwahati	Guwahati	Assam

- person(ID, name, email)

Creating table:

```
CREATE TABLE person (
      ID         INT              PRIMARY KEY,
      name       VARCHAR(50)      NOT NULL,
      email      VARCHAR(50)      UNIQUE     NOT NULL
);
```

Inserting data:

```
INSERT INTO person VALUES
(1, 'Arnav Singhal', 'arnav.singhal@example.com'),
(2, 'Priya Sharma', 'priya.sharma@example.com'),
(3, 'Rohan Mehta', 'rohan.mehta@example.com'),
(4, 'Neha Patil', 'neha.patil@example.com'),
(5, 'Karan Iyer', 'karan.iyer@example.com'),
(6, 'Sneha Reddy', 'sneha.reddy@example.com'),
(7, 'Aditya Verma', 'aditya.verma@example.com'),
(8, 'Ishita Das', 'ishita.das@example.com'),
(9, 'Rahul Nair', 'rahul.nair@example.com'),
(10, 'Meera Joshi', 'meera.joshi@example.com');
```

Table contents:

```
+----+---------------+---------------------------+
| ID | name          | email                     |
+----+---------------+---------------------------+
|  1 | Arnav Singhal | arnav.singhal@example.com |
|  2 | Priya Sharma  | priya.sharma@example.com  |
|  3 | Rohan Mehta   | rohan.mehta@example.com   |
|  4 | Neha Patil    | neha.patil@example.com    |
|  5 | Karan Iyer    | karan.iyer@example.com    |
|  6 | Sneha Reddy   | sneha.reddy@example.com   |
|  7 | Aditya Verma  | aditya.verma@example.com  |
|  8 | Ishita Das    | ishita.das@example.com    |
|  9 | Rahul Nair    | rahul.nair@example.com    |
| 10 | Meera Joshi   | meera.joshi@example.com   |
+----+---------------+---------------------------+
```

- programme(progId, title, fromdate, todate)

Creating table:

```
CREATE TABLE programme (
      progId      INT               PRIMARY KEY,
      title       VARCHAR(100)      NOT NULL,
      fromdate    DATE              NOT NULL,
      todate      DATE              NOT NULL,
      CHECK (fromdate < todate)
);
```

Inserting data:

```
INSERT INTO programme VALUES
(1, 'AI and Machine Learning Workshop', '2025-01-10', '2025-01-15'),
(2, 'Data Science Bootcamp', '2025-02-05', '2025-02-12'),
(3, 'Cloud Computing Fundamentals', '2025-03-01', '2025-03-05'),
(4, 'Cybersecurity Awareness Week', '2025-04-10', '2025-04-14'),
(5, 'Web Development Internship Orientation', '2025-05-20', '2025-
05-25'),
(6, 'IoT Applications Training', '2025-06-03', '2025-06-07'),
(7, 'Blockchain for Beginners', '2025-07-15', '2025-07-20'),
(8, 'Quantum Computing Seminar', '2025-08-01', '2025-08-03');
```

Table contents:

```
+--------+---------------------------------------+------------+------------+
| progId | title                                 | fromdate   | todate     |
+--------+---------------------------------------+------------+------------+
|      1 | AI and Machine Learning Workshop      | 2025-01-10 | 2025-01-15 |
|      2 | Data Science Bootcamp                 | 2025-02-05 | 2025-02-12 |
|      3 | Cloud Computing Fundamentals          | 2025-03-01 | 2025-03-05 |
|      4 | Cybersecurity Awareness Week          | 2025-04-10 | 2025-04-14 |
|      5 | Web Development Internship Orientation | 2025-05-20 | 2025-05-25 |
|      6 | IoT Applications Training             | 2025-06-03 | 2025-06-07 |
|      7 | Blockchain for Beginners              | 2025-07-15 | 2025-07-20 |
|      8 | Quantum Computing Seminar             | 2025-08-01 | 2025-08-03 |
+--------+---------------------------------------+------------+------------+
```

```
-- Verifying that the check constraint works (this query fails as
fromdate > todate)
INSERT INTO programme VALUES
(9, 'Python Workshop', '2025-10-17', '2025-10-03');
```

Output:

```
ERROR 3819 (HY000): Check constraint 'programme_chk_1' is violated.
```

- coordinator(ID, progId, centreId)

Creating table (assuming one person can coordinate more than one program):

```
CREATE TABLE coordinator (
      ID         INT,
      progId     INT,
      centreId   INT,
      PRIMARY KEY (ID, progId, centreId),
      FOREIGN KEY (ID) REFERENCES person(ID),
      FOREIGN KEY (progId) REFERENCES programme(progId),
      FOREIGN KEY (centreId) REFERENCES remotecentre(centreId)
);
```

Inserting data:

```
INSERT INTO coordinator (ID, progId, centreId) VALUES
(1, 1, 1),
(2, 2, 3),
(3, 3, 2),
(4, 4, 4),
(5, 5, 6),
(6, 6, 5),
(7, 7, 8),
(8, 8, 7),
(1, 2, 1),
(3, 4, 2),
(5, 7, 6),
(6, 8, 5);
```

Table contents:

```
+----+--------+----------+
| ID | progId | centreId |
+----+--------+----------+
|  1 |      1 |        1 |
|  1 |      2 |        1 |
|  2 |      2 |        3 |
|  3 |      3 |        2 |
|  3 |      4 |        2 |
|  4 |      4 |        4 |
|  5 |      5 |        6 |
|  6 |      6 |        5 |
|  5 |      7 |        6 |
|  7 |      7 |        8 |
|  6 |      8 |        5 |
|  8 |      8 |        7 |
+----+--------+----------+
```

- participant(ID, progId, centreId)

  Creating table (assuming one person can participate in more than one program):

```
CREATE TABLE participant (
      ID         INT,
      progId     INT,
      centreId   INT,
      PRIMARY KEY (ID, progId, centreId),
      FOREIGN KEY (ID) REFERENCES person(ID),
      FOREIGN KEY (progId) REFERENCES programme(progId),
      FOREIGN KEY (centreId) REFERENCES remotecentre(centreId)
);
```

  Inserting data:

```
INSERT INTO participant VALUES
(2, 1, 1),
(3, 1, 1),
(4, 2, 3),
(10, 5, 6),
(1, 5, 6),
(2, 6, 5),
(4, 6, 5),
(5, 7, 8),
(6, 7, 8),
(7, 8, 7),
(9, 8, 7);
```

  Table contents:

```
+----+--------+----------+
| ID | progId | centreId |
+----+--------+----------+
|  2 |      1 |        1 |
|  3 |      1 |        1 |
|  4 |      2 |        3 |
|  1 |      5 |        6 |
| 10 |      5 |        6 |
|  2 |      6 |        5 |
|  4 |      6 |        5 |
|  5 |      7 |        8 |
|  6 |      7 |        8 |
|  7 |      8 |        7 |
|  9 |      8 |        7 |
+----+--------+----------+
```